Ex. no. 5

POS Tagging and Document Similarity Using Spacy and TF-IDF

AIM:

To perform POS tagging using spacy & find the most relevant documents by comparing a query with given documents using TF-IDF and cosine similarity

PROCEDURE:

1. POS tagging: Use spacy to tag parts of speech
2. Vectorization: Convert documents & query into numerical vectors
3. Similarity: Compute cosine similarity
4. Ranking: Display documents in order of similarity score.

PROGRAM:

```
import spacy
from sklearn.feature_extraction.text import TfidVectorizer
from sklearn.metrices.paircoise import cosine_similarity.
nlp=spacy.load("en_core_web_sm")
text="AI-driven platforms personalize learning paths
      for students."
for token in nlp(text):
    print(f"{token.text:<12}->{token.pos_}")

documents = [
    "AI tools analyze student performance & provide
real-time feedback",
```

```
    "AI helps automate grading & administrative
    tasks in schools.",
    "Chatbots assist students with questions
    anytime.",
    "Virtual classrooms powered by AI enhance
    engagement."
]
query = "How does AI support students in learning?"
corpus = documents + [query]
vectorizer = TfidVectorizer()
tfidf = vectorizer.fit_transform(corpus)
similarities = cosine_similarity(tfidf[-1], tfidf[:-1])
.flatten()
ranked = sorted(zip(similarities, documents), reverse=True)
print("\nTop relevant documents:")
for score, doc in ranked:
    print(f"Score: {score:.2f} -> {doc}")
```

RESULT:

Thus, POS tagging using spacy & document similarity using TF-IDF & cosine similarity were successfully implemented & executed.

OUTPUT:

AI-driven -> ADJ
platforms -> NOUN
personalize -> VERB
learning -> NOUN
paths -> NOUN
for -> ADP
students -> NOUN
. -> PUNCT

Top relevant documents:

Score: 0.65 -> AI tools analyze student performance & provide real-time feedback.